

DATE: ___/___/___

DAY: ___/___/___

: Sections: A

MCQs:

- 1) A
- 2) $\sqrt{5}$ (B)
- 3) C
- 4) ~~A~~ B
- 5) A
- 6) D
- 7) C
- 8) D
- 9) B
- 10) D
- 11) A
- 12) C

Section - C

Q: 3(i)

A: 3(i)

$$(a) (n+1)(n+2)(n+3)(n+4) - 24$$

$$\begin{aligned}
 & (n+2)(n+3)(n+1)(n+4) - 24 \\
 &= [n(n+3) + 2(n+3)][n(n+4) + 1(n+4)] - 24 \\
 &= (n^2 + 3n + 2n + 6)(n^2 + 4n + n + 4) - 24 \\
 &= (n^2 + 5n + 6)(n^2 + 5n + 4) - 24
 \end{aligned}$$

$$\text{Let } n^2 + 5n = a$$

$$\begin{aligned}
 & (a+6)(a+4) - 24 \\
 &= a(a+4) + 6(a+4) - 24 \\
 &= a^2 + 4a + 6a + 24 - 24 \\
 &= a^2 + 10a \\
 &= a(a+10) \\
 &= (x^2+5x)(x^2+5x+10) \\
 &= x(x+5)(x^2+5x+10)
 \end{aligned}$$

(b) $4x^2 + 12xy + 9y^2$

$$\begin{aligned}
 & (2x)^2 + 2(2x)(3y) + (3y)^2 \\
 &= (2x+3y)^2
 \end{aligned}$$

Q: 3(ii)

A: 3(ii)

(a) Any three logarithm rules:-

(1) $\log mn = \log m + \log n$

(2) $\log_m \frac{m}{n} = \log m - \log n$

(3) $\log m^n = n \log m$

(b) i) $\log 42$

$$\log(6 \times 7)$$

$$\log(2 \times 3 \times 7)$$

$$\log 2 + \log 3 + \log 7$$

$$0.3010 + 0.4771 + 0.8451$$

$$= 1.6232$$

$$\textcircled{ii} \quad \log \frac{9}{2}$$

$$\log \frac{3^2}{2}$$

$$\begin{aligned} & \log 3^2 - \log 2 \\ & 2 \log 3 - \log 2 \\ & 2(0.4771) - 0.3010 \\ & 0.9542 - 0.3010 \\ & 0.6532 \end{aligned}$$

$$\text{————— } Q: 23 \text{ (iii) —————}$$

$$\text{————— } A: 3 \text{ (iii) —————}$$

$$\textcircled{a} \quad \frac{n-2}{3^2} + \frac{n+1}{2} = 4$$

$$\frac{(n-2)^{x_2}}{3^{x_2}} + \frac{(n+1)^{x_3}}{2^{x_3}} = 4$$

$$\frac{2n-4}{6} + \frac{3n+3}{2} = 4$$

$$\frac{2n+3n-4+3}{6} = 4$$

$$\frac{5n-1}{6} = 4 \quad (\text{cross multiplication})$$

(b) $3x - 5 \leq 7, x \in \mathbb{R}$

$$3x \leq 7 + 5$$

$$3x \leq 12$$

$$x \leq \frac{12}{3}$$

$$x \leq 4$$

$$x \leq 4$$

$$\{x | x \in \mathbb{R}, x \leq 4\}$$



Q.3(iv)

A.3(iv)

$A(-4, 2), B(1, 4), C(3, -1), D(-2, 3)$

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

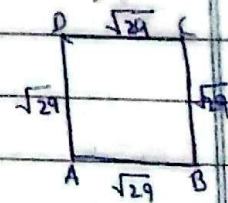
$$= \sqrt{(1 - (-4))^2 + (4 - 2)^2}$$

$$= \sqrt{(1+4)^2 + (2)^2}$$

$$= \sqrt{(5)^2 + 4}$$

$$= \sqrt{25 + 4}$$

$$= \sqrt{29}$$



$$d_{BC} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3 - 1)^2 + (-1 - 4)^2}$$

$$= \sqrt{(2)^2 + (-5)^2}$$

$$= \sqrt{4 + 25}$$

$$= \sqrt{29}$$

$$\begin{aligned}
 d_{10} &= \frac{\sqrt{(-2-3)^2 + (-3-(-1))^2}}{\sqrt{(-5)^2 + (-2)^2}} \\
 &= \frac{\sqrt{25+4}}{\sqrt{29}}
 \end{aligned}$$

$$\begin{aligned}
 d_{10} &= \frac{\sqrt{(-2-(-4))^2 + (-3-2)^2}}{\sqrt{(2+4)^2 + (-5)^2}} \\
 &= \frac{\sqrt{(-2)^2 + 25}}{\sqrt{4+25}} \\
 &= \frac{\sqrt{29}}{\sqrt{29}}
 \end{aligned}$$

formula of distance is:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Section-B

Q: 2(i) (or) ———
A: 2(i) (or) ———

(a) $\sqrt{36}$
= 6 (rational, perfect square)

(b) $\sqrt{10}$
(irrational, not a perfect square)

(c) -0.7
= $-\frac{7}{10}$ (rational, can change into fraction)

Q: 2(ii) (or)

A: 2(ii) (or)

(a) 0.00046
 $= 4.6 \times 10^{-4}$

(b) $92,000,000$
 $= 9.2 \times 10^7$

Q: 2(iii) (or)

A: 2(iii) (or)

$$\log \frac{a^2 b}{c}$$

$$\log a^2 + \log b - \log c$$

$$2 \log a + \log b - \log c$$

Q: 2(iv)

A: 2(iv)

(a) $\log 6$
 $\log(2 \times 3)$
 $\log 2 + \log 3$
 $0.3010 + 0.4771$
 0.7781

(b) $\log 12$
 $\log(4 \times 3)$
 $\log(2^2 \times 3)$
 $\log 2^2 + \log 3$

$$\begin{aligned}
 & 2 \log 2 + \log 3 \\
 & 2(0.3010) + 0.4771 \\
 & 0.602 + 0.4771 \\
 & = 1.0791
 \end{aligned}$$

$$\text{Q: } 2(v) \text{ —————}$$

$$\text{A: } 2(v) \text{ —————}$$

(a)

$$\begin{aligned}
 & x^2 - 9x + 20 \\
 & x^2 - 4x - 5x + 20 \\
 & x(x-4) - 5(x-4) \\
 & (x-5)(x-4)
 \end{aligned}$$

(b)

$$\begin{aligned}
 & 2x^2 + 7x + 3 \\
 & 2x^2 + 6x + x + 3 \\
 & 2x(x+3) + 1(x+3) \\
 & (2x+1)(x+3)
 \end{aligned}$$

$$\text{Q: } 2(vi) \text{ —————}$$

$$\text{A: } 2(vi) \text{ —————}$$

$$12x^2y^3, 18x^3y^2$$

$$12 = 2^2 \times 3 \quad 18 = 3^2 \times 2$$

$$x^2 = \textcircled{x} \times \textcircled{x} \quad x^3 = x \times \textcircled{x} \times \textcircled{x}$$

$$y^3 = y \times \textcircled{y} \times \textcircled{y} \quad y^2 = \textcircled{y} \times \textcircled{y}$$

$$\text{HCF} = 6x^2y^2$$

Q.2(vii)

A.2(vii)

$$\frac{2x}{3} + 1 = \frac{x}{2} + 2$$

$$\frac{2x^{x1}}{3^{x1}} + \frac{1 \times 3}{1 \times 3} = \frac{x^{x1}}{2^{x1}} + \frac{2^{x2}}{1^{x2}}$$

$$\frac{2x + 3}{3} \xrightarrow{\text{cross multiply}} \frac{x + 4}{2} \quad (\text{cross multiply})$$

$$3(x+4) = 2(2x+3)$$

$$3x + 12 = 4x + 6$$

$$3x - 4x = +6 - 12$$

$$-x = -6$$

$$x = 6$$

Q.2(viii)

A.2(viii)

$$8x + 9 < 6x - 7$$

$$8x - 6x < -7 - 9$$

$$2x < -16$$

$$x < -16^8$$

$$x < -8$$

$$\{x \mid x \in \mathbb{R}, x < -8\}$$

Q.2 (in)

A: 2 (in)

A(-2, 1) & B(1, 5)

$$d_{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(1 - (-2))^2 + (5 - 1)^2}$$

$$= \sqrt{(1+2)^2 + (4)^2}$$

$$= \sqrt{(3)^2 + 16}$$

$$= \sqrt{9 + 16}$$

$$\sqrt{25} \Rightarrow 5 \text{ units.}$$

Q.2 (in)

A: 2 (in)

$$\text{midpoint of } AB = \frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2}$$

$$= \frac{6 + 2}{2}, \frac{9 + 5}{2}$$

$$= \frac{8}{2}, \frac{14}{2}$$

$$= 4, 7$$

Q.2 (in)

A: 2 (in)

Chord: A line segment whose two endpoints both lie on the circle.

DATE: ___/___/___

DAY: ___/___/___

Tangent :-

A straight line touches the circle at exactly one point. (the point of contact)

Arc :- The continuous part of the circumference (curved boundary) ~~is~~ of the circle.